

Building and Evaluating an AI Customer-Support Agent for Spotify from Noisy Twitter Conversations

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Abstract

We build an AI customer-support agent for a single brand (Spotify) from the *Customer Support on Twitter* dataset. The agent classifies each incoming message into eight data-derived intents, drafts a reply grounded in the brand’s own past resolutions via retrieval, and decides whether to auto-handle or escalate to a human with a stated reason. Consistent with the assignment’s emphasis that “the proof is worth more than the system,” the focus is a rigorous evaluation: a hand-labelled golden set, two baselines, an LLM-as-judge validated against human scores, and a documented failure analysis. On a fixed 81-example test split, a few-shot LLM classifier reaches **0.588 macro-F1**, beating a TF-IDF logistic-regression baseline (0.391) by +0.20 and a majority-class baseline (0.035) decisively. We report where the headline number is misleading and what one additional week of work would target.

1 Problem Framing

We select Spotify (*SpotifyCares*) from data: among five high-volume brands it is the most *bounded* product (yielding a clean intent set) while still having a genuinely balanced escalate-vs-auto-handle signal ($\sim 31\%$ of replies push the customer to a private channel), so neither decision class is degenerate.

What “good” means here. Spotify’s Twitter support is high-volume, low-stakes, and repetitive. A good agent should (i) route each message to the right intent, (ii) draft replies *grounded* in how Spotify has actually resolved similar issues rather than inventing advice, and (iii) above all *know when to escalate* — a wrong auto-reply on a billing dispute is far costlier than an unnecessary escalation. We therefore optimise escalation for precision on auto-handling, accepting some over-escalation.

What we did not build. No multi-turn dialogue management (we act on the first customer message), no live account integration (anything needing a private lookup is escalated, not faked), and no fine-tuning (few-shot only, for reproducibility and cost).

2 System Overview

The pipeline is: reconstruct conversation threads $\rightarrow$ derive an intent taxonomy $\rightarrow$ classify $\rightarrow$ retrieve similar resolved threads $\rightarrow$ draft a grounded reply $\rightarrow$ decide escalate/auto-handle, followed by an evaluation layer (baselines, LLM-as-judge, failure analysis). We reconstruct 28,277 Spotify threads from the reply graph; of these, $\sim 17.8k$ are resolved in-thread and form the retrieval corpus, while $\sim 10.5k$ “take-it-to-DM” threads double as weak escalation labels.

3 Intent Taxonomy

Rather than guessing intents, we cluster the first customer message of every thread (TF-IDF + k -means) to surface natural groups, then author a small eight-label set: `playback_app.issue`, `account.access`, `billing.payment`, `subscription.management`, `content.availability`, `playlist.help`, `praise.or.feedback`, and `other.unclear`. Two deliberate choices: we split `billing.payment` from `subscription.management` because they behave oppositely for triage (a charge dispute needs an account lookup and escalates; a cancellation can be answered with instructions), and we include two non-support intents because a large share of brand mentions are not problems — recognising “no action needed” is itself good triage.

4 Experimental Setup

Golden set. We hand-label 202 examples using a hybrid sampling scheme: a stratified portion guarantees every intent has enough examples for a real per-class F1, and a random portion preserves the true class distribution. The two answer different questions (per-class capability vs. honest real-world performance).

Split and metric. A single fixed stratified split (121 train / 81 test) is used by every method, so comparisons are apples-to-apples; few-shot demonstrations are drawn only from train. The headline metric is *macro-F1*, which weights rare intents equally so a model cannot win by

only serving the majority class.

5 Results

Table 1 reports intent-classification performance on the fixed test split. The LLM classifier beats the TF-IDF baseline by **+0.20 macro-F1**. The gains concentrate exactly where a lexical model is weakest: **account_access** ($0.29 \rightarrow 0.61$ — it reads “hacked, weird emails” as a security issue without the literal token “password”) and **other_unclear** ($0.00 \rightarrow 0.67$ — TF-IDF cannot represent “no clear intent applies,” whereas the LLM reasons about it).

Escalation / auto-handle. On a 40-message agent run, 19 messages were auto-handled with a grounded reply and 21 were escalated, with escalations concentrated in the account-specific intents a human should own (billing (5), account access (4), subscription/family plan (2), plus complex playback cases). Retrieval was strong (embedding similarity 0.6–1.0 on matched cases), so most escalations were driven by the reply-grounding check rather than weak retrieval. Escalate-decision performance vs. the “did Spotify DM?” weak label ($n=40$): **precision 0.62, recall 0.87, F1 0.72, accuracy 0.75** — the agent catches 87% of the cases Spotify actually escalated, at the cost of some over-escalation, the conservative trade-off we intended.

Reply quality. An LLM-as-judge scores auto-handled replies 1–5 on a grounded/correct/helpful/on-brand rubric; judge–human agreement is measured with quadratic-weighted Cohen’s κ on 18 hand-scored replies. **Mean judge score 4.47/5**; judge–human agreement **exact 67%, within-1 78%**, $\kappa = 0.445$ (moderate). The moderate κ is itself informative: the judge is directionally reliable but should not be trusted to a decimal — a near-perfect κ would more likely mean the human rubber-stamped the judge than that the judge is flawless. Representative grounded outputs: a hacked-account message is routed to account-recovery steps; a “when is [album] coming?” message is answered “not available yet, try the singles,” matching Spotify’s real historical reply; an album-art bug is met with Spotify’s standard device/OS/version triage opener.

6 Failure Analysis

Both the TF-IDF and LLM classifiers fail in the *same* boundary regions, which indicates genuine label ambiguity rather than model-specific quirks.

1. **billing_payment vs. subscription_management.** e.g. “charged for premium subscription via mobile provider bill” — a charge and a subscription action co-occur; even human labellers hesitate.
2. **praise_or_feedback vs. playlist_help.** Feature suggestions (“you should let me put podcasts in playlists”)

carry topic keywords that pull lexical models toward the wrong intent; the speech act matters more than the topic.

3. **playback_app_issue is a weak-signal catch-all.** “All my albums disappeared. App is updated. Help” overlaps library/account symptoms; it is the lowest-F1 class.
4. **account_access symptoms read as playback.** Security framing (“account hacked, weird emails”) is missed without login/password tokens — an area where the LLM improves over TF-IDF.
5. **other_unclear bleeds into everything** for lexical models ($F1 = 0$ on the split), since “unclear” is defined by the *absence* of a clear intent; the LLM recovers this to 0.67.

7 What Is Misleading About the Headline Number?

- **Tiny test set.** TF-IDF scores 0.391 on the split but 0.517 ± 0.074 under 5-fold CV; an 81-item split hides a ± 0.07 swing, so any headline macro-F1 should be read as ± 0.05 –0.07.
- **Some “errors” are label ambiguity.** The boundary confusions above are cases where the true label is debatable; macro-F1 penalises them as harshly as egregious misses, understating real usefulness.
- **The LLM number is from a *mix* of models.** A free-tier cap of 20 requests/day/model forced rotation across Gemini Flash variants, so the score estimates “Flash-class” capability, not a single model.
- **Escalation rate is inflated by quota.** In the agent run the daily cap ran out mid-way, so some messages escalated for “drafting unavailable” rather than on a policy decision; the reported escalation split is an upper bound.
- **Escalation ground truth is a heuristic.** “Did Spotify DM them?” is a noisy proxy for “should this escalate.”
- **The judge is one model grading another,** trustworthy only to the extent of the reported human-agreement κ .

8 Engineering Notes

Real free-tier constraints shaped the system: every LLM response is cached to disk (re-runs cost no quota and are reproducible), calls are rate-limited and retried on connection resets / 503s, and the client rotates across models on a per-model daily cap. Retrieval uses local sentence-embeddings with an automatic TF-IDF fallback, keeping it off the API and resilient to a constrained network.

Table 1: Intent classification, macro-F1 on the fixed 81-example test split.

Method	macro-F1	accuracy
Trivial (majority class)	0.035	0.160
Simple (TF-IDF + LogReg)	0.391	0.432
TF-IDF, 5-fold CV (robustness)	0.517 $\pm$ 0.074	—
LLM few-shot (Gemini Flash)	0.588	0.580

9 What I Would Do With One More Week

Grow the golden set to ~ 600 with a second annotator and report inter-annotator agreement (shrinking the confidence intervals); add a confidence threshold that routes low-confidence intents to escalation; resolve the ambiguous boundaries via a sub-type field or targeted labelled examples; replace the DM-heuristic escalation label with a small hand-labelled escalation set; and run the classifier on a single model to remove the mixed-model caveat.

A Decision Log (selected)

1. Optimised for evaluation over system, per the brief.
2. Chose Spotify from data (bounded product, balanced escalation, legible).
3. Mined the taxonomy by clustering rather than guessing.
4. Split billing from subscription (opposite triage behaviour).
5. Included two non-support intents to keep the classifier honest.
6. Hybrid golden-set sampling (stratified for per-class + random for truth).
7. One fixed train/test split for all methods; demos from train only.
8. Macro-F1 as headline metric (class imbalance).
9. Two baselines as the bar, not throwaways.
10. Local embeddings for retrieval with TF-IDF fallback.
11. Grounding corpus = resolved threads; DM threads = weak escalation labels.
12. Conservative escalation with a stated reason on every decision.
13. Cached + rate-limited + model-rotating LLM client for the free tier.
14. Validated the judge against human scores via quadratic-weighted κ .

Data and Reproducibility

Dataset: *Customer Support on Twitter* (Kaggle, [thoughtvector/customer-support-on-twitter](https://www.kaggle.com/thoughtvector/customer-support-on-twitter)). A subsample is used throughout, as the brief encourages. A README reproduces the headline results in under 15 minutes.